2. Lecture 6

Part A due Mar 6, 2024 20:30 IST Completed

The following can be done after Lecture 6.

6-1

5.0/5.0 points (graded)

Let $\mathbf{D} := \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. What is the upper right entry of $e^{t\mathbf{D}}$?

0

✔ Answer: 0

Solution:

0.

In general, if $\mathbf{D}$ is a diagonal matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$, then $e^{t\mathbf{D}}$ is a diagonal matrix with diagonal entries $e^{\lambda_1 t}, \ldots, e^{\lambda_n t}$. The upper right entry of $e^{t\mathbf{D}}$ in the problem is not on the diagonal, so it is 0.

Submit You have used 1 of 10 attempts

Answers are displayed within the problem

6-2

5.0/5.0 points (graded)

Let $\mathbf{A} = \begin{pmatrix} 2 & 2 \\ -2 & -2 \end{pmatrix}$. What is the lower right entry of $e^{t\mathbf{A}}$? (Hint: What is $\mathbf{A}^2$?)

1-2*t

✔ Answer: 1-2*t

$1 - 2 \cdot t$

? FORMULA INPUT HELP

Solution:

$1 - 2t$.

Since $\mathbf{A}^2 = 0$,

$$\begin{aligned} e^{t\mathbf{A}} &= \mathbf{I} + t\mathbf{A} + \frac{(t\mathbf{A})^2}{2!} + \frac{(t\mathbf{A})^3}{3!} + \cdots \\ &= \mathbf{I} + t\mathbf{A} \\ &= \begin{pmatrix} 1+2t & 2t \\ -2t & 1-2t \end{pmatrix}. \end{aligned}$$

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Answers are displayed within the problem

6-3

10.0/10.0 points (graded)

Calculator

Let $\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 0 & 3 \end{pmatrix}$. What is the upper right entry of $e^{t\mathbf{A}}$?

2*t*e^(3*t)

$2 \cdot t \cdot e^{3 \cdot t}$

Answer: 2*t*e^(3*t)

Solution:

$2te^{3t}$.

Let $\mathbf{N} := \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$. Then $\mathbf{A} = 3\mathbf{I} + \mathbf{N}$. Scalar multiples of $\mathbf{I}$ commute with any matrix, and $\mathbf{N}^2 = 0$, so

$$\begin{aligned} e^{t\mathbf{A}} &= e^{t(3\mathbf{I})} e^{t\mathbf{N}} \\ &= e^{3t\mathbf{I}} \left(\mathbf{I} + t\mathbf{N} + 0 + 0 + \cdots\right) \\ &= \begin{pmatrix} e^{3t} & 0 \\ 0 & e^{3t} \end{pmatrix} \begin{pmatrix} 1 & 2t \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} e^{3t} & 2te^{3t} \\ 0 & e^{3t} \end{pmatrix}. \end{aligned}$$

Submit

You have used 1 of 15 attempts

Answers are displayed within the problem

6-4

10.0/10.0 points (graded)

Find the first coordinate of $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$ with respect to the basis $\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ of $\mathbb{R}^2$.

-11

Answer: -11

Solution:

The first coordinate is -11.

We need to solve

$$\begin{pmatrix} 5 \\ 7 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 3 \\ 2 \end{pmatrix};$$

then the answer is c_1. This amounts to the system

$$\begin{aligned} 2c_1 + 3c_2 &= 5 \\ c_1 + 2c_2 &= 7. \end{aligned}$$

To find c_1, eliminate c_2 by taking 2 times the first equation minus 3 times the second equation:

$$c_1 = 2(5) - 3(7) = -11.$$

Submit

You have used 3 of 10 attempts

Answers are displayed within the problem

Calculator

Previous Next

6-5

10/10 points (graded)

The function $2\cos(5t) + 10\sin(5t)$ lies in the complex vector space with basis e^{5it}, e^{-5it}. Find its second coordinate with respect to that basis.

Solution:

The answer is $1+5i$, because

$$\begin{aligned}2\cos(5t)+10\sin(5t) &= 2\left(\frac{e^{5it}+e^{-5it}}{2}\right)+10\left(\frac{e^{5it}-e^{-5it}}{2i}\right)\\ &= 1\left(e^{5it}+e^{-5it}\right)-5i\left(e^{5it}-e^{-5it}\right)\\ &= (1-5i)\,e^{5it}+(1+5i)\,e^{-5it}.\end{aligned}$$

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Answers are displayed within the problem

6-6

5.0/5.0 points (graded)

The vectors $\mathbf{v}_1 := \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}$ and $\mathbf{v}_2 := \begin{pmatrix} -3/5 \\ 4/5 \end{pmatrix}$ form an orthonormal basis of $\mathbb{R}^2$. Find the first coordinate of $\begin{pmatrix} 10 \\ 15 \end{pmatrix}$ with respect to this basis.

17

Answer: 17

Solution:

The first coordinate is found by taking the dot product

$$\begin{pmatrix} 10 \\ 15 \end{pmatrix}\cdot \mathbf{v}_1 = \begin{pmatrix} 10 & 15 \end{pmatrix}\begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix} \quad (4.185)$$

$$= 8 + 9 = 17. \quad (4.186)$$

Submit You have used 2 of 10 attempts

Answers are displayed within the problem

6-7

10/10 points (graded)

Vectors $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ form an orthogonal basis for $\mathbb{R}^3$. Given that $\mathbf{w}_1 = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$, what is the first coordinate of the vector $\mathbf{v} := \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ with respect to this basis?

Calculator

(Enter as a fraction or decimal to four places.)

13/38

Answer: 13/38

$\frac{13}{38}$

Solution:

The first coordinate is $13/38$.

If $\mathbf{v} = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + c_3\mathbf{w}_3$, then $\mathbf{v}\cdot\mathbf{w}_1 = c_1\mathbf{w}_1\cdot\mathbf{w}_1 + 0 + 0$ (since $\mathbf{w}_1$ is orthogonal to $\mathbf{w}_2$ and $\mathbf{w}_3$), so the first coordinate is

$$c_1 = \frac{\mathbf{v}\cdot\mathbf{w}_1}{\mathbf{w}_1\cdot\mathbf{w}_1} = \frac{\begin{pmatrix}0 & 1 & 2\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}}{\begin{pmatrix}2 & 3 & 5\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}} = \frac{13}{38}.$$

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You have used 6 of 10 attempts

Answers are displayed within the problem

2. Lecture 6

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